

L2.7 Newton's Method

Recall: L2.6b More Optimization

Name _____

Solo – 9 minutes

1. A continuous function f has exactly one critical number c on the open interval $(0, \infty)$. To the left of c , $f' < 0$; to the right of c , $f' > 0$. What may you conclude about $f(c)$, and what is the theorem called?

2. Find the point on the parabola $y = x^2$ that is closest to the point $(0, 3)$.

Do not solve it. Write the optimization equation, the constraint, and the interval.

3. Find the roots of $x^2 - 10x + 21$.

Now find the roots of $x^5 - x - 1 = 0$.